

Article

Security Challenges in 5G Network Slicing: A Risk-Based Analysis and Conceptual Framework

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Abstract

Network slicing is a core enabler of multi-tenant 5th Generation (5G) architectures, allowing heterogeneous services to coexist over shared infrastructure. However, ensuring effective isolation between slices remains a critical security challenge, as failures may enable cross-slice interference, data leakage, or cascading service disruption. This article analyses security vulnerabilities affecting 5G network slicing from a risk-oriented perspective, with particular emphasis on isolation weaknesses across orchestration, virtualization, network, and interface layers. Due to the technical immaturity and instability of current open-source slicing platforms, experimental validation of security mechanisms proved infeasible. These limitations are therefore treated as empirical evidence informing a structured vulnerability taxonomy and a qualitative risk assessment grounded in confidentiality, integrity, and availability. Building on this analysis, the article proposes a conceptual security framework that integrates defence-in-depth, zero-trust principles, continuous monitoring, and adaptive response mechanisms to enforce isolation dynamically. Aligned with established standards and regulatory references, the framework provides a coherent theoretical foundation for future experimental validation and the secure design of resilient 5G network slicing architectures.

Keywords: 5G network slicing; slice isolation; security framework; risk-based analysis; orchestration security; multi-tenant networks

1. Introduction

5G mobile networks represent a fundamental shift in the design and operation of communication infrastructures, enabling the support of highly heterogeneous services with diverse performance, latency, and reliability requirements. Among the architectural innovations introduced by 5G, network slicing has emerged as a key enabler for multi-tenant environments, allowing multiple network slices to coexist over a shared physical infrastructure while being tailored to specific use cases such as enhanced mobile broadband, massive Internet of Things (IoT), and ultra-reliable low-latency communications.

While network slicing provides unprecedented flexibility and scalability, it also introduces significant security challenges. In particular, ensuring effective isolation between slices is critical to prevent unauthorized access, resource leakage, and cross-slice interference in multi-tenant scenarios. Failures in slice isolation may allow attacks originating in one slice to propagate to others, undermining service guarantees and compromising trust in 5G infrastructures [1,2].



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Existing research has extensively explored security threats in 5G network slicing, including vulnerability analyses, attack taxonomies, and mitigation strategies targeting different architectural layers. Previous studies—including our earlier work—have contributed valuable insights into potential attack vectors and theoretical risks [3,4]. However, most existing approaches remain fragmented, often focusing on isolated components or abstract threat models, with limited consideration of integrated risk assessment and actionable security frameworks aligned with realistic 5G architectures [5].

This reveals a critical gap in the current state of the art: the absence of comprehensive, risk-oriented approaches that systematically assess slice isolation threats and translate them into coherent security frameworks applicable to practical 5G deployments. In particular, there is a lack of conceptual models that connect identified vulnerabilities, architectural constraints, and mitigation mechanisms within a unified analytical structure.

To address this gap, this article proposes a risk-oriented analytical approach to 5G network slicing security and introduces a conceptual security framework aimed at strengthening slice isolation. The main contributions of this work are threefold:

- A structured synthesis of security challenges affecting slice isolation, grounded in a refined vulnerability taxonomy;
- The definition of a risk-oriented framework that integrates isolation, monitoring, and response mechanisms across slicing-related architectural layers;
- A set of practical guidelines to support future research and the secure design of 5G network slicing environments.

This article is organized as follows. Section 2 provides an overview of the related work on 5G network slicing, including the technological background and a synthesis of existing security analyses and mitigation approaches. Section 3 discusses the limitations encountered in experimental validation using current open-source 5G slicing platforms, motivating the adoption of a theoretical and risk-oriented analytical approach. Section 4 presents a structured, risk-based analysis of vulnerabilities that affect slice isolation across different architectural layers. In Section 5, a risk-oriented security framework is proposed, integrating isolation-by-design principles, automation, and continuous monitoring. Section 6 discusses the proposed framework in light of the existing literature, highlighting practical implications and limitations. Finally, Section 7 concludes the article and outlines directions for future research.

2. Overview of the Related Work

This section provides an overview of existing research on 5G network slicing with a particular focus on security and slice isolation. It first introduces the technological and architectural background that underpins network slicing, including virtualization and orchestration mechanisms commonly adopted in both research and experimental settings. Then it synthesizes the main strands of the literature addressing security models, vulnerability analyses, and mitigation strategies, highlighting the limitations of current approaches and motivating the need for an integrated, risk-oriented perspective.

2.1. Background

Network slicing is a fundamental capability of 5G architectures, enabling the creation of multiple network slices over a shared physical infrastructure, each tailored to specific service requirements and operating within multi-tenant environments. This paradigm is made possible by the adoption of virtualization and programmability principles, notably Software Defined Networking (SDN) [6], Network Functions Virtualization (NFV) [7], and Management and Orchestration (MANO) frameworks. Together, these technologies

support the dynamic instantiation, scaling, and management of slice-specific resources across the control, user, and management planes.

In research and experimental contexts, several open-source platforms have been adopted to explore 5G network slicing concepts. European Telecommunications Standards Institute (ETSI) Open Source MANO (OSM) is commonly used for NFV orchestration and lifecycle management, while Free5GC provides an open implementation of the 5G Core network, and UERANSIM enables simulation of user equipment and radio access components. These platforms play an important role in academic research by offering transparency and accessibility, allowing researchers to study slicing mechanisms, orchestration workflows, and architectural interactions.

However, despite their relevance, open-source platforms exhibit structural limitations when compared to carrier-grade deployments. Slice support is often limited to Single Network Slice Selection Assistance Information (S-NSSAI), with restricted enforcement of resource or data-plane isolation. Orchestration capabilities remain immature, and security mechanisms are typically implemented as add-ons rather than being embedded by design. These constraints limit the realism of experimental studies and highlight the gap between conceptual slicing models and their practical realization, a challenge that motivates further analytical investigation.

2.2. State of the Art

The existing literature on 5G network slice security covers a broad range of theoretical models, vulnerability analyses, and mitigation strategies. A significant body of work proposes taxonomies to classify threats and security requirements across slicing architectures, often focusing on NFV infrastructures, orchestration layers, and multi-tenant environments. These taxonomies provide valuable conceptual frameworks for understanding complex threat surfaces, particularly those related to inter-slice isolation, resource sharing, and orchestration misconfigurations [8–10].

Several studies further analyze known vulnerabilities and attack vectors that affect network slicing, highlighting risks such as cross-slice data leakage, insecure management interfaces, weak authentication mechanisms, and misconfigured orchestration policies. These works consistently identify slice isolation as a critical security requirement spanning multiple architectural layers, from virtualized infrastructure to control and management planes. Despite this, most analyses remain largely theoretical or rely on simulation-based evaluations, with limited validation in realistic slicing environments [11–14].

To address these challenges, a growing number of contributions explore mitigation strategies based on security-by-design principles and intelligent automation. Recent approaches incorporate Artificial Intelligence and Machine Learning techniques for anomaly detection, adaptive resource allocation, and proactive threat mitigation. While these solutions demonstrate promising results in controlled or simulated settings, they are typically developed as proof-of-concept frameworks and face significant integration challenges when applied to current open-source or operational platforms [15–17].

In summary, the state of the art reveals a fragmented landscape: while theoretical models, taxonomies, and intelligent mitigation techniques are well developed, their alignment with practical slicing architectures remains limited. The lack of integrated, risk-oriented approaches that connect architectural vulnerabilities, implementation constraints, and actionable security mechanisms constitutes a key gap in existing research. This gap underpins the analytical focus adopted in this work and motivates the structured vulnerability analysis and framework proposal presented in the following sections [18–20].

3. Experimental Research Methodology

The initial phase of the research underlying this work was conceived with an experimental focus, aiming to assess slice isolation and security vulnerabilities through the deployment of multi-slice 5G environments. The original objective was to simulate inter-slice attack scenarios and evaluate mitigation mechanisms using open-source platforms, in line with prevailing approaches in experimental network security research.

Open-source tools were deliberately selected due to their accessibility, transparency, and widespread adoption in academic studies. Platforms such as ETSI OSM, Free5GC, and UERANSIM are frequently used to explore 5G architectures, orchestration workflows, and slicing concepts without the constraints imposed by proprietary solutions. In principle, these tools offer the flexibility required to investigate slice configuration, traffic separation, and control-plane behavior in controlled environments.

However, the experimental validation phase, conducted up to the second quarter of 2025, revealed significant technical and architectural limitations that directly affected the feasibility of security-focused experimentation. A primary constraint was the absence of effective slice isolation. While logical slice identifiers are supported, current open-source platforms do not enforce physical or data-plane separation between slices, preventing meaningful evaluation of inter-slice attack vectors or isolation failures. As a result, all slices effectively share the same transport and processing resources, undermining the core assumptions required for slice-level security testing.

In addition to isolation limitations, the experimental environments exhibited a high degree of instability. Recurrent component failures, version incompatibilities, and fragile inter-dependencies between network functions compromised the consistency of test executions. These issues significantly reduced the reproducibility of experiments, a critical requirement for empirical security research. Furthermore, the limited visibility and control over internal mechanisms—such as traffic handling, policy enforcement, and security instrumentation—restricted the ability to observe, measure, or validate security-relevant events.

Table 1 summarizes the experimental results obtained with the selected platforms, highlighting their operational status, principal limitations, and relevance to the research objectives. Rather than reflecting isolated implementation issues, these results reveal structural constraints that affect the suitability of current open-source tools for rigorous security experimentation in multi-slice environments.

Table 1. Summary of experimental findings and limitations of open-source 5G slicing platforms.

Tool	Test Outcome	Main Limitations	Relevance to Research
OSM (v16)	Not functional	Requires SOL006 packages; lack of available VNF support	Unusable for slice orchestration and isolation testing
Free5GC (v3.3.0)	Partially functional	Platform instability; absence of real slice isolation mechanisms	Limited applicability for cross-slice attack simulation
UERANSIM (v3.2.6)	Partially functional	Version incompatibilities; unstable integration with Free5GC	Insufficient as a standalone environment for security evaluation

Collectively, these constraints had a substantial impact on the research process. The inability to reliably instantiate isolated slices, execute adversarial scenarios, or reproduce results rendered the experimental approach unsuitable for rigorous security evaluation. Importantly, these limitations are not incidental implementation issues, but rather reflect broader maturity gaps in current open-source 5G slicing platforms.

In light of these findings, the research methodology was deliberately reframed. Instead of pursuing partial or unreliable experiments, this work adopts a theoretical and analytical approach grounded in a risk-oriented perspective. The observed experimental limitations are treated as empirical evidence informing the analysis of architectural vulnerabilities, guiding the development of a structured vulnerability taxonomy and a conceptual security framework. This shift ensures scientific rigor while directly addressing the gap between theoretical security models and the practical readiness of contemporary 5G network slicing environments.

4. Detailed Vulnerability Analysis

Building on the limitations identified in the experimental validation, this section presents a structured and risk-oriented vulnerability analysis for slicing the 5G network. Rather than examining isolated weaknesses, the analysis adopts a systemic perspective, focusing on how vulnerabilities emerge from the interaction between architectural layers, shared resources, and management mechanisms inherent to multi-tenant slicing environments.

The analysis is grounded in a multi-layered vulnerability taxonomy previously introduced in earlier work and refined in the context of this study [3]. The taxonomy captures the main attack surfaces affecting slice isolation and reflects the operational lifecycle of network slices across the 5G architecture. Specifically, vulnerabilities are classified according to four architectural layers:

- Orchestration and management layer;
- Virtualization layer;
- Network layer;
- Interface and communication layer.

This layered classification provides a structured analytical lens for understanding how security weaknesses manifest at different levels of the slicing stack. More importantly, it enables the identification of dependencies between layers, which play a central role in amplifying the impact of individual vulnerabilities. In network slicing architectures, orchestration decisions, shared infrastructure, and service-based interfaces are tightly coupled. As a result, failures or misconfigurations in one layer may propagate across others, leading to systemic risks that extend beyond the initially compromised component.

To support prioritization, vulnerability analysis follows a qualitative risk-oriented approach grounded in the Confidentiality, Integrity, and Availability (CIA) triad. Each vulnerability category is assessed based on its potential to compromise these fundamental security properties, taking into account the shared and multi-tenant nature of slicing environments. Particular emphasis is placed on availability and propagation effects, as disruptions affecting shared components can simultaneously degrade multiple slices and tenants, resulting in large-scale service impact.

From this perspective, the analysis highlights a set of architectural domains that consistently concentrate the most critical risks:

- Centralised orchestration and management functions, which control slice lifecycle operations and influence all downstream components;
- Shared virtualized infrastructure and physical resources, where vulnerabilities can directly undermine isolation guarantees and enable lateral movement across slices;
- Exposed Application Programming Interface (API) and control interfaces, which govern slice instantiation, modification, and termination, and significantly expand the attack surface.

The orchestration and management layer emerges as a dominant source of systemic risk due to its central role in controlling slice lifecycle operations and shared resource allocation. Vulnerabilities related to resource management, security misconfigurations, and insufficiently protected management APIs can enable unauthorized slice manipulation, privilege escalation, or large-scale denial-of-service conditions. Given the centralized role of orchestration platforms, exploitation at this layer often produces cascading effects that are difficult to contain and may compromise multiple slices simultaneously. As illustrated in Figure 1, orchestration functions act as a control nexus between interface-level requests, virtualized infrastructure, and slice lifecycle workflows; consequently, a malicious or misconfigured operation affecting a single slice—such as excessive scaling requests or policy violations—can propagate through shared control logic, triggering resource contention, misallocation, or service disruption across other co-hosted slices.

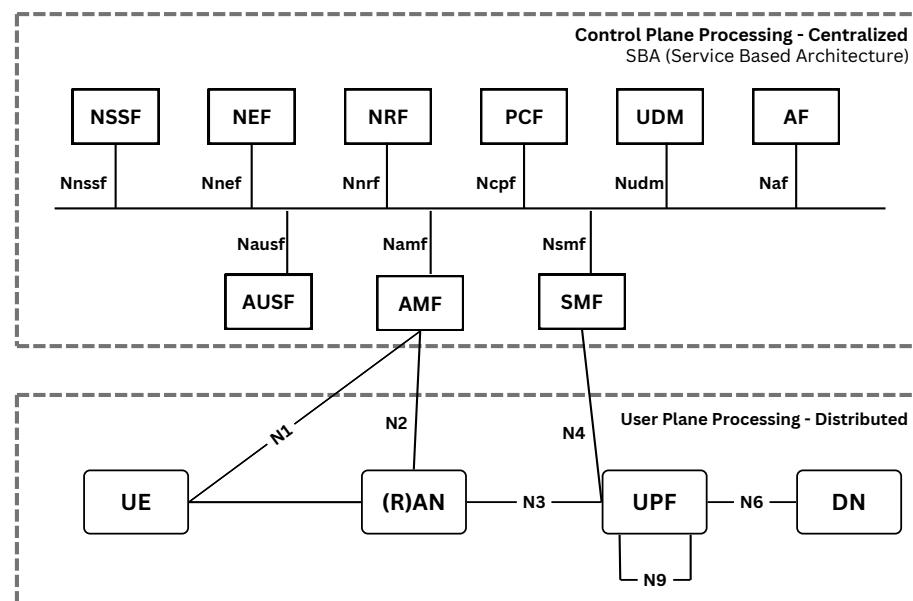


Figure 1. 5G core architecture, highlighting the centralized control plane processing and distributed user plane processing components.

The virtualization layer represents another critical risk domain. Network functions belonging to different slices frequently coexist in shared compute, storage, and networking infrastructure. Vulnerabilities affecting hypervisors, container runtimes, or physical hosts therefore pose a direct threat to slice isolation. Successful exploitation may allow attackers to bypass logical separation mechanisms, move laterally between slices, or disrupt availability across multiple services. These risks illustrate the limitations of relying solely on logical isolation without strong enforcement at the infrastructure level.

At the network layer, weak enforcement of slice isolation constitutes a non-negligible security risk. In many implementations, slice separation relies predominantly on logical identifiers, such as the S-NSSAI, and configuration parameters, without providing strict guarantees of traffic isolation or effective resource segregation between coexisting slices. When shared user plane components or routing elements are not slice-aware, this may result not only in cross-slice interference and performance degradation, but also in unintended data exposure, directly affecting confidentiality in multi-tenant environments. In this context, data-plane confidentiality mechanisms—such as encryption applied directly to traffic payloads—become a relevant defence-in-depth measure, ensuring that sensitive information remains protected even if logical isolation boundaries are partially bypassed [21].

Finally, the interface and communication layer exposes a broad attack surface through service-based interfaces and management APIs. Weak authentication mechanisms, excessive privileges, or inadequate protection of control-plane communications can facilitate unauthorized access, interception of sensitive data, and escalation of privileges across slice boundaries. As API architectures rely heavily on API-driven interactions, vulnerabilities at this layer frequently constitute effective entry points for broader systemic compromise.

Overall, this risk-oriented analysis demonstrates that vulnerabilities affecting slice isolation do not arise from isolated design flaws but from the combined effects of architectural complexity, shared resources, and insufficient enforcement mechanisms across multiple layers. The cumulative impact of inadequately mitigated vulnerabilities significantly increases systemic risk in 5G multi-tenant environments. These findings provide a clear rationale for prioritizing security controls and directly inform the design principles underpinning the security framework proposed in the following section.

5. Conceptual Framework for Security Solutions

Building upon the vulnerability taxonomy and risk assessment presented in the previous section, this section introduces a risk-oriented conceptual framework designed to enhance security and isolation in 5G network slicing environments. While the earlier analysis identified critical attack surfaces and assessed their potential impact across architectural layers, the framework proposed here translates those findings into a coherent security architecture that integrates prevention, monitoring, and adaptive response mechanisms.

The framework is intentionally conceptual rather than implementation-specific. Its objective is to provide a structured foundation that can guide the design, evaluation, and future validation of secure slicing platforms, while remaining aligned with realistic deployment constraints and current standardization efforts.

5.1. Framework Scope and Architecture

The proposed framework addresses the systemic nature of security risks in multi-tenant 5G slicing by explicitly linking vulnerabilities to security requirements and mitigation mechanisms. It adopts a layered architectural perspective, consistent with the slicing taxonomy introduced earlier, and emphasizes the cumulative effects of vulnerabilities that propagate across orchestration, virtualization, network, and interface domains.

The primary design goals of the framework are to:

- Reinforce slice isolation as a core architectural property;
- Enable continuous visibility over security-relevant events;
- Support automated and adaptive response to incidents;
- Ensure coherence with recognized standards and regulatory requirements.

Rather than focusing on individual controls, the framework promotes an integrated security posture in which preventive, detective, and responsive mechanisms operate in a coordinated manner.

5.2. Guiding Security Principles

The framework is grounded in a set of guiding principles that reflect both established security best practices and the specific characteristics of 5G slicing architectures:

- Defence-in-depth, ensuring that multiple, independent security controls are deployed across architectural layers to limit the impact of single-point failures;
- Zero-trust slicing, whereby no slice, network function, or management component is implicitly trusted, and all interactions are subject to continuous authentication and authorization;

- Isolation by design, treating isolation as a fundamental architectural requirement rather than a side effect of virtualization technologies;
- Adaptive automation, which allows security mechanisms to respond dynamically to anomalies and evolving threat conditions.

Together, these principles shape the placement and interaction of security mechanisms throughout the slicing lifecycle, from slice instantiation to runtime operation and recovery.

5.3. Multi-Layer Framework Architecture

The framework follows a multi-layer architecture that mirrors the functional structure of 5G network slicing. Security mechanisms are distributed across the orchestration and management layer, the virtualization layer, the network layer, and the interface and communication layer, while a cross-cutting monitoring and response capability operates across all domains.

At a high level, the framework is organized around three interacting functional domains:

- Preventive controls, which enforce isolation, access control, and secure configuration at each layer;
- Monitoring and detection, which collect and correlate telemetry data to identify deviations from expected behavior;
- Automated response and recovery, which enable timely containment and mitigation of detected incidents.

These domains form a closed feedback loop, allowing security policies and enforcement mechanisms to adapt continuously based on observed conditions. This feedback-driven design is illustrated in Figure 2, which is retained from the thesis to support conceptual clarity. While the framework incorporates well-established security mechanisms, its contribution does not lie in the novelty of individual controls, but in their integration into a coherent, risk-driven and feedback-oriented architecture that enables adaptive enforcement of slice isolation across layers.

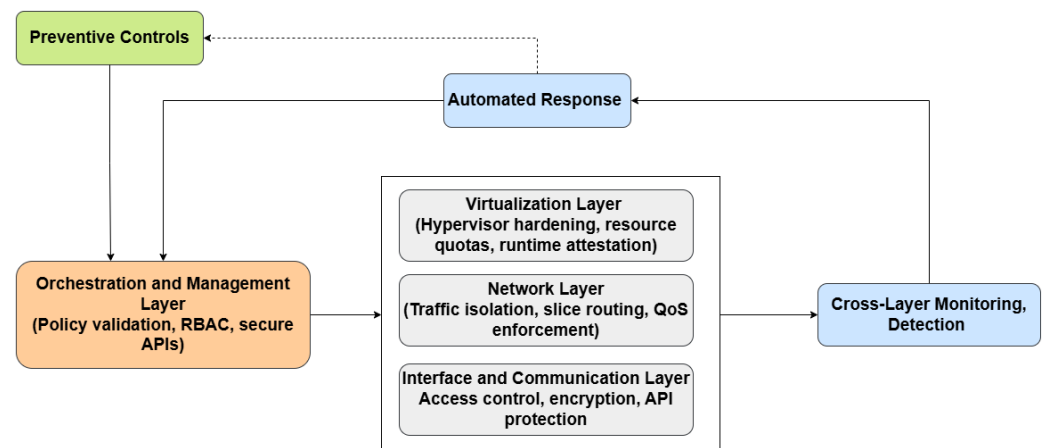


Figure 2. Feedback-driven security architecture for 5G network slicing. The flow illustrates how cross-layer monitoring triggers automated responses, how these responses update orchestration policies, and how preventive controls are continuously refined through this feedback loop.

5.4. Mapping Vulnerabilities to Security Mechanisms

A central contribution of the proposed framework lies in the explicit and systematic mapping between the vulnerability categories identified in the risk-oriented analysis and the corresponding security mechanisms designed to mitigate them. Rather than addressing threats in isolation, the framework ensures that each architectural layer contributes

complementary controls that collectively reduce the likelihood, impact, and propagation of cross-slice attacks.

At the orchestration and management layer, security mechanisms primarily protect centralized control-plane functions and slice lifecycle operations, whose compromise may affect multiple slices through shared orchestration logic. Hardened management APIs, strict role-based access control, automated policy validation for slice descriptors, comprehensive audit logging, and request throttling are employed to mitigate risks such as orchestrator overload, misconfiguration exploitation, and unauthorized slice modification.

Within the virtualization layer, the framework reinforces isolation between co-hosted network functions through a combination of hypervisor and container hardening, hardware-assisted virtualization, slice-level resource quota, and runtime integrity verification. These measures directly address threats related to lateral movement, privilege escalation, and shared resource abuse, effectively reducing the blast radius of compromises affecting the underlying infrastructure.

At the network layer, slice-aware traffic segmentation, micro-segmentation policies, encryption of intra-slice communication, and continuous traffic monitoring are introduced to prevent cross-slice interference, data leakage, and resource exhaustion. These controls ensure that the logical separation between slices is consistently enforced in the data plane, even in the presence of shared transport components.

Finally, at the interface and communication layer, exposed attack surfaces are mitigated by using secure communication protocols, API gateway protections, strong authentication mechanisms, and continuous vulnerability assessment of external interfaces. These controls limit the risk of eavesdropping, command injection, and unauthorized access originating from external or inter-domain interactions.

Table 2 summarizes the relationship between the proposed security mechanisms and major 5G standards and regulatory frameworks, including 3rd Generation Partnership Project (3GPP) [22], ETSI NFV-MANO [23], National Institute of Standards and Technology (NIST) SP 800-207 [24], Network and Information Systems Directive 2 (NIS2) [25], and European Union Agency for Cybersecurity (ENISA) [26] recommendations. The table provides a condensed view of how the framework operationalization recognized best practices while remaining aligned with both technical specifications and regulatory obligations.

Rather than assuming a uniform architectural model, the proposed framework reconciles these standardization perspectives by abstracting common security functions and control objectives. This approach allows mechanisms defined under differing architectural assumptions—such as telco-centric management models, virtualization-oriented orchestration, and risk-driven governance frameworks—to be aligned at a functional and risk-governance level.

To further clarify how the vulnerability-to-mitigation mapping presented above operates in practice, the following narrative illustrates the end-to-end handling of a representative cross-slice attack within the proposed framework.

Consider the attack scenario “Slice Flooding via Orchestrator Overload”. During the design time, this vulnerability may arise from overly permissive slice descriptors, insufficient validation of lifecycle requests, or the absence of effective rate constraints on management APIs. Under these conditions, the orchestrator becomes a critical attack surface due to its centralized control over slice instantiation and scaling, where excessive requests can overwhelm control-plane resources and indirectly affect multiple tenants sharing the infrastructure.

Within the proposed framework, preventive controls at the orchestration and interface layers aim to reduce the likelihood of this attack through hardened management APIs, strict role-based access control, request throttling, and automated policy validation of slice

descriptors. If preventive measures are bypassed or misconfigurations persist, cross-layer monitoring continuously correlates orchestration metrics, API request rates, and system load indicators. This correlated telemetry enables the detection of abnormal request patterns and resource consumption trends indicative of orchestrator overload.

Table 2. Mapping of security mechanisms across 5G standards and regulatory frameworks.

Framework Mechanism	3GPP SA5 (OAM Management)	ETSI NFV-MANO	NIST ZTA/NIS2/ENISA 5G
Identity assurance; MFA; OAM federation	OAM access control; NRM inventory	Authentication policies; SOL API security	Continuous identity verification; identity governance; credential and access control
Micro-segmentation; ABAC policies	Slice profiles; domain separation	Isolation via VIM/VNFM/Orchestrator; connectivity rules	Dynamic attribute-based access; segmentation; lateral movement mitigation
Slice isolation; firewalling	Slice lifecycle control; isolation KPIs	NS/NSD isolation constraints; forwarding domains	Micro-perimeters; network resilience; virtualization risk mitigation
API hardening (mTLS, rate limiting, authz)	Secure OAM and service interfaces	SOL API logging, authentication and integrity	Least-privilege enforcement; policy monitoring; API exposure limitation
Telemetry; logging; anomaly detection	FM/PM alarms and performance metrics	Observability; lifecycle events; VNF/NS metrics	Continuous monitoring; SOC operations; threat detection
Vulnerability management; patching	Change management; maintenance windows	VNF/VM/container updates; catalogue maintenance	Secure baselines; vulnerability remediation obligations
Image integrity; supply-chain assurance	Artifact validation; version compliance	Signed onboarding; trusted catalogues	Trust verification; supplier vetting; supply-chain risk mitigation
Configuration compliance; drift control	CM/NRM auditing; consistency checks	Declarative NS/NSD compliance policies	Policy-as-code; governance; secure configuration validation
Incident response; reporting	Rollback procedures; FM/PM correlation	Automation for isolation and redeployment	Incident reporting; coordinated response workflows
Business continuity; disaster recovery	Slice-level redundancy and resilience	Multi-site orchestration; backup mechanisms	Operational continuity; service resilience measures

Once such anomalous behavior is detected, the feedback-driven response logic activates orchestration-level mitigation actions according to predefined risk thresholds and slice criticality. These actions may include dynamic rate limiting of management requests, temporary isolation or suspension of the offending slice, and rollback of recent configuration changes. Through this closed-loop interaction between monitoring and orchestration, the framework enables adaptive containment of the attack while preserving service continuity for unaffected slices and maintaining overall system stability.

Beyond standards alignment, this mapping highlights an essential property of the proposed framework: each critical vulnerability category is mitigated through a combination of preventive, detective, and responsive controls spanning multiple architectural layers.

This cross-layer coverage reduces single points of failure and ensures that weaknesses not detected at one layer can be identified or contained at another. As a result, the framework does not merely catalogue countermeasures, but establishes a coherent security logic that links risk assessment, architectural design, and governance requirements into a unified model for secure 5G network slicing.

5.5. Monitoring, Detection, and Automated Response

Recognizing the limitations of purely preventive security controls, the framework incorporates a cross-layer monitoring and automated response subsystem. This subsystem collects telemetry from orchestration platforms, virtualization environments, network components, and interfaces, enabling correlated analysis of security-relevant events.

Anomalies are classified according to their scope and severity, distinguishing between performance anomalies, configuration deviations, and security incidents. Based on this classification, predefined response policies are triggered, including slice isolation, policy rollback, resource reallocation, and credential revocation.

By integrating detection and response mechanisms directly with the orchestration layer, the framework supports adaptive security governance and reduces both detection and response times. This feedback-driven approach transforms the framework into a dynamic security model capable of maintaining isolation and availability even under active attack conditions. In scenarios where an automated security response—such as slice isolation—may conflict with service continuity requirements (e.g., potential Service Level Agreement (SLA) violations for critical slices), the framework deliberately defers precedence decisions to implementation-level policies, allowing operators to define response priorities based on slice criticality, risk severity, and operational context.

Within this adaptive response logic, the framework explicitly acknowledges slicing-specific trade-offs, such as strict isolation versus multiplexing efficiency or slice elasticity versus policy stability. Rather than enforcing fixed responses, security actions are governed by policy-driven thresholds that take into account slice criticality, risk severity, and operational context. As a result, more or less stringent isolation and mitigation measures may be dynamically applied, balancing security enforcement and service continuity through continuous monitoring and orchestration-driven decisions.

Although the framework does not quantify the performance overhead of runtime attestation, this overhead is inherently dependent on several implementation specific variables. These include the choice of cryptographic primitives (e.g., hash functions and integrity mechanisms), attestation frequency, execution context (inline versus off-path), and traffic load on high-throughput user plane functions. These factors are therefore identified as key dimensions for future experimental evaluation.

5.6. Theoretical Validation and Standards Alignment

Although the proposed framework has not been empirically validated, its robustness is supported by a structured theoretical validation grounded in coverage and consistency analysis. Coverage analysis confirms that each category of vulnerability identified in the proposed taxonomy is addressed by at least one preventive control and one adaptive or responsive mechanism, ensuring that no critical attack surface remains unmitigated. Consistency analysis further verifies that the interaction between automation, orchestration, and enforcement mechanisms does not introduce conflicting dependencies that could weaken isolation guarantees or undermine system stability.

Beyond internal coherence, the framework was designed to align with established technical standards and regulatory references relevant to 5G network slicing. These include 3GPP specifications for 5G management and operations (SA5), ETSI NFV-MANO

architectural principles, the NIST SP 800-207 Zero Trust Architecture, the European NIS2 Directive, and the ENISA 5G Security Toolbox. This alignment ensures that the proposed mechanisms are not only theoretically sound, but also compatible with realistic operational, governance, and compliance requirements.

To complement the standards-oriented perspective, Table 3 provides a concise validation view by mapping representative cross-slice attack scenarios to the primary mitigation mechanisms and the architectural layers involved. This table is derived from the more extensive mappings presented in the dissertation and is retained here to demonstrate systematic threat coverage while preserving the conciseness required in an article format.

Table 3. Mitigation mechanisms applied to representative cross-slice attack scenarios.

Attack Scenario	Primary Mitigation Mechanisms	Layer(s) Involved
Slice Flooding via Orchestrator Overload	Rate limiting, request throttling, API authentication, load balancing of orchestration services	Orchestration and Management
Misconfiguration Exploitation in Slice Deployment	Policy validation, NSD/VNFD template verification, configuration consistency checks	Orchestration and Management
Unauthorized Slice Modification via API Abuse	RBAC enforcement, mutual TLS, audit logging, token-based access control	Orchestration and Management, Interface
Hypervisor Escape Leading to Cross-Slice Compromise	Hypervisor hardening, hardware-assisted virtualization, runtime integrity verification	Virtualization
Targeted Hardware-Induced DoS Affecting All Slices	Resource quotas, hardware-level monitoring, adaptive VNF migration, redundancy mechanisms	Virtualization, Monitoring
Unauthorized Data Access via Shared Routing or Misconfigured Interfaces	Traffic segmentation (VLAN/VXLAN), SDN micro-segmentation, encrypted intra-slice communication	Network
Cross-Slice Resource Exhaustion (CPU or Bandwidth Contention)	Slice-level quotas, dynamic resource allocation, anomaly-based performance monitoring	Virtualization, Network
Passive Traffic Sniffing Across Slices	TLS/IPsec enforcement, secure logging, interface-level telemetry auditing	Interface and Communication, Network
Unauthorized Command Injection into Slice Control Functions	Input validation, API gateway filtering, multi-factor authentication, credential revocation on detection	Interface and Communication, Orchestration

The mapping confirms that attacks targeting centralized orchestration functions, shared virtualized infrastructure, and inter-slice communication paths—where failures may propagate across multiple slices—are consistently mitigated through layered controls spanning prevention, detection, and response. Notably, several attack scenarios are addressed by mechanisms operating across multiple layers, reinforcing the defence-in-depth and zero-trust principles underlying the framework. In this context, Zero Trust decision logic is realized through continuous monitoring metrics and policy-driven thresholds embedded in the feedback-driven architecture, enabling dynamic policy evaluation and enforcement analogous to Policy Decision Points within the 5G MANO environment.

Overall, the theoretical validation demonstrates that the framework achieves both horizontal coverage across attack categories and vertical coherence across architectural layers. By combining standards alignment with explicit vulnerability-to-mitigation mapping, the framework establishes a credible and extensible foundation for future experimental validation and for the design of security-aware 5G slicing platforms in operational contexts.

5.7. Summary

This section presented a risk-oriented conceptual framework for enhancing security and isolation in 5G network slicing environments. By systematically translating vulnerability analysis into architectural principles, layered security mechanisms, and adaptive monitoring and response capabilities, the framework provides coherent theoretical coverage of the main attack vectors affecting multi-tenant slicing systems.

The framework establishes a structured foundation for future experimental validation and for the design of secure orchestration and isolation mechanisms in practical 5G deployments. The following section discusses the implications of these findings, as well as the limitations and future research directions arising from this work.

6. Analysis and Discussion of Results

This section critically analyzes the proposed security framework in light of vulnerability assessment and the current state of research on 5G network slicing security. The discussion focuses on the interpretation of the findings, the positioning of the framework relative to existing approaches, and its practical implications and limitations.

6.1. Interpretation of Findings and Framework Architecture

The vulnerability and risk analysis demonstrated that the most critical threats to slice isolation emerge primarily from the orchestration and virtualization layers. These layers concentrate control over resource allocation, slice lifecycle management, and shared execution environments, making them natural amplification points for cascading failures across slices.

High-severity scenarios—such as orchestrator overload, misconfiguration exploitation, and cross-slice resource exhaustion—highlight that isolation breaches are rarely confined to individual components. Instead, they propagate through shared control logic and infrastructure dependencies, blurring the boundaries between confidentiality, integrity, and availability. This observation reinforces the need to treat slice isolation as a dynamic property that must be continuously enforced and verified.

The conceptual framework proposed in this work was explicitly designed as a response to these systemic risks. By combining defence-in-depth, zero-trust principles, and adaptive automation, the framework shifts isolation from a static configuration assumption to an actively monitored and enforced security condition. The integration of preventive controls with continuous monitoring and automated response mechanisms provides a coherent theoretical basis for mitigating both direct attacks and cascading effects.

6.2. Positioning with Respect to the State of the Art

Existing research on 5G network slicing security has produced valuable contributions in areas such as attack taxonomies, AI-assisted orchestration, and zero-trust access control. However, these efforts are often fragmented, addressing individual layers or mechanisms without an integrated risk-driven perspective.

Compared to the state of the art, the primary contribution of the proposed framework lies in the systematic integration of three elements that are rarely combined:

- An architecture-aware vulnerability taxonomy;

- A risk-oriented prioritization of attack vectors;
- A multi-layer mitigation logic incorporating prevention, detection, and response.

While several studies explore anomaly detection or intelligent orchestration, they frequently lack explicit traceability between detected events, assessed risk, and concrete mitigation actions. The framework presented here addresses this gap by explicitly mapping vulnerabilities to security mechanisms and embedding adaptive response logic within the orchestration process itself.

Rather than introducing novel security primitives, the framework reorganizes existing mechanisms into a coherent, risk-oriented and slice-aware architecture. This positions it as a reference model capable of complementing and extending current approaches aligned with 3GPP, ETSI NFV-MANO, and zero-trust design principles.

6.3. Practical Implications and Limitations

From a practical perspective, the framework offers guidance for operators, network architects, and researchers working with multi-tenant 5G slicing environments. It highlights the critical role of orchestration and virtualization layers in enforcing effective isolation and underscores the need for tighter integration between lifecycle management, monitoring, and automated response.

The model also supports compliance-oriented risk assessment by providing traceability between attack vectors, CIA impact, and mitigation strategies—an increasingly relevant requirement under regulatory frameworks such as the NIS2 Directive and ENISA 5G Security Toolbox.

Nevertheless, the framework remains theoretically validated. Due to the technical instability and functional limitations of current open-source slicing platforms, empirical validation was not feasible within the scope of this work. Future research should therefore focus on implementing and evaluating the proposed architecture in controlled experimental environments to assess scalability, performance overhead, and detection accuracy.

Looking beyond current 5G deployments, the transition toward the Sixth Generation (6G) is expected to further increase the complexity of managing shared spectrum and infrastructure resources. Emerging paradigms such as the use of upper mid-band spectrum introduce new coexistence, coverage, and capacity trade-offs, which may amplify isolation, orchestration, and governance challenges in multi-tenant environments [27]. In this sense, the maturity gaps observed in current 5G slicing platforms can be seen not only as present limitations, but also as early indicators of the security and management challenges anticipated in future 6G architectures.

The adaptive and cross-layer nature of the framework inevitably introduces additional complexity and computational overhead. However, this overhead is predictable and proportional to the security benefits achieved. In multi-tenant slicing environments, where isolation failures can trigger cascading service disruptions, the trade-off between complexity and resilience is both necessary and justified.

While this work does not provide a quantitative performance evaluation, such overhead is an important consideration for practical deployments. In a post-implementation context, isolation overhead may be assessed using simple operational metrics, such as the relative increase in end-to-end latency introduced by slice isolation mechanisms:

$$\text{Isolation Overhead} = \frac{\text{Latency}_{\text{with isolation}} - \text{Latency}_{\text{baseline}}}{\text{Latency}_{\text{baseline}}} \quad (1)$$

The metric presented in Equation (1) is provided as an illustrative example to guide future experimental assessments and is not used for risk scoring or validation within the scope of this article.

7. Conclusions and Future Work

This article examined the security challenges associated with 5G network slicing, focusing on isolation weaknesses in multi-tenant environments. Building upon the underlying dissertation, the work consolidated a risk-oriented vulnerability analysis and proposed a conceptual framework aimed at strengthening isolation and architectural resilience.

The analysis showed that the most critical isolation failures stem from shared orchestration and virtualization components, where weak policy enforcement, misconfigurations, and resource contention may propagate across slices. By introducing a cross-layer vulnerability taxonomy aligned with the 5G architectural model and a risk-based severity assessment, the study provided a structured approach to identifying and prioritizing slicing-specific threats.

In response, a conceptual security framework was proposed, treating isolation as a dynamic and continuously enforced property rather than a static configuration. The framework integrates defence-in-depth, zero-trust principles, and adaptive monitoring and response mechanisms across all slicing layers. Its alignment with established standards and regulatory references, including 3GPP SA5, ETSI NFV-MANO, NIST SP 800-207, NIS2, and the ENISA 5G Security Toolbox, reinforces its relevance to both research and operational contexts.

Although empirical validation was not feasible due to current limitations of open-source slicing platforms, this constraint highlights the gap between architectural security requirements and their practical implementation. Future work should therefore focus on experimental validation in controlled testbeds, the integration of intelligent and policy-based orchestration, the use of advanced Artificial Intelligence (AI)/Machine Learning (ML) techniques for adaptive security, and the extension of the proposed approach to emerging 6G scenarios.

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